

Software Requirements Specification (SRS)

Afia — AI-Driven Dietary Health Assistant

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1 User Stories & Rationale

Each user story below is derived directly from recurring pain points identified across 900+ negative Google Play reviews of MyFitnessPal, Lose It!, and Yazio (April 2026 dataset).

1.1 US-01: Arabic-First Experience

As a new user (including Arabic-speaking users),

I want a welcome / language-selection screen to appear before any sign-up or log-in screen, where the copy and layout of that first screen follow my device locale — Arabic when the device is set to Arabic, English when the device is set to English — while I can still explicitly choose the other language for the app; after I confirm my preferred language, I want all authentication and onboarding screens, food data, recipes, and AI prompts to follow that choice; and after I am signed in, I want to change the language at any time from Settings with my preference persisted across sessions, **so that** I am not forced through English-only auth flows on an Arabic-locale device, I can track my nutrition entirely in my chosen language without external translators, and I retain full control to switch languages later.

Solves: Language Barrier

Evidence: `arabic_language_support` is the #1 issue signal in Arabic reviews across all three competitors (MFP signal=305, LoseIt signal=57, Yazio signal=36). A single MFP review requesting Arabic received 156 upvotes. Users explicitly state they delete the app solely because of the absence of Arabic.

1.2 US-02: Flexible Meal Logging & Discovery

As a calorie-conscious user,

I want multiple ways to add food to my diary: primarily by photographing my plate for an AI-generated calorie estimate, but also by logging meals manually, searching my previously logged meals, or requesting personalized meal suggestions aligned with my dietary plan, **so that** I can log meals rapidly with a photo when possible, easily find or enter exact nutritional details when the AI cannot recognize the food, and discover new healthy options without leaving the logging flow.

Solves: Logging Friction & Lack of Inspiration

1.3 US-03: Accurate & Correctable Nutritional Data

As a health-tracking user,

I want AI-generated calorie estimates I can manually override before or after saving, **so that** my food log remains accurate even when the AI estimation is imprecise or I need to correct an accidental entry.

Solves: Inaccurate Data & Correction Barrier

Evidence: A MFP user reported calorie counts for “1 cup cooked quinoa” ranging from 120 to 225 calories in the same search results (crowd-sourced data). A Yazio user wrote: “My weight is wrong forever” after being unable to edit an accidental entry. `search_logging` signals (which include data accuracy complaints) exceeded 100 in all three apps.

1.4 US-04: Local & Arab Cuisine Awareness

As a user from an Arab country,

I want the recipe converter and snack suggestions to recognize traditional Arab dishes (e.g., *koshari*, *fattoush*, *molokhia*),

so that I can track my actual daily meals instead of substituting Western food items that do not match what I eat.

Solves: Western-Centric Food Database

Evidence: Arabic-language reviews across all three competitors request local food coverage. High-frequency Arabic complaint terms include transliterated terms such as *koshari* and *al-aklat al-sha’biyya* (popular/traditional dishes). One MFP Arabic review (215 upvotes) explicitly asked for “popular Arab dishes” and conditioned a 5-star rating on it.

1.5 US-05: Rapid, Error-Free Onboarding

As a first-time user,

I want to complete my profile setup — including weight, height, activity level, and dietary goals — within 3 minutes with proper input validation,

so that I am not blocked by crashes, frozen screens, or unit-conversion errors during my first interaction with the app.

Solves: First-Minute Failure / Broken Onboarding

Evidence: `login_opening_issues` appears as an issue theme in Arabic reviews for MFP (signal=53), Yazio (signal=7), and LoseIt (signal=5). A MFP Arabic user (150 upvotes) reported: “The app stops at entering weight and does not respond.” `performance_bugs` signals exceed 50 across all three competitors in English reviews.

1.6 US-06: Information-Dense Progress Dashboard

As a long-term user tracking my dietary progress,

I want a dashboard that shows my calorie intake, macros, and weight trend on a single scrollable view using charts and compact grids,

so that I can assess my progress without navigating through multiple screens of decorative padding.

Solves: UI Regression / Information Density Loss

Evidence: `ux_navigation` is a recurring English issue theme for MFP (signal=105), LoseIt (signal=62), and Yazio (signal=46). A LoseIt user noted: “I can only see two food choices” after an

update added redundant buttons and oversized cards. A Yazio user described needing “6 taps to exit one menu.”

1.7 US-07: Healthy Alternative Discovery

As a user with dietary restrictions or health goals,
I want to receive AI-generated healthy recipe alternatives and snack suggestions tailored to my profile,
so that I can discover nutritious options without external research or guesswork.

Solves: Lack of Actionable Guidance

Evidence: Competitors focus on passive tracking (log what you ate) but offer no proactive nutritional guidance within the free tier. Users across all three apps describe the experience as “just a calorie counter” with no meaningful health insights. Yazio review (76 upvotes): “Worthless... all this is, is a fancy calorie counter.”

1.8 US-08: Water Intake Tracking

As a health-conscious user,
I want to log my daily water intake and monitor my hydration progress alongside calories and nutrition,
so that I can maintain a more complete picture of my health habits rather than tracking food alone.

Solves: Incomplete Health Tracking

Evidence: Competing apps focus heavily on calorie counting but do not consistently treat hydration as a first-class tracking feature within the main experience. This creates a gap for users who want a unified wellness dashboard that includes both nutrition and water intake.

2 Functional Requirements

2.1 US-01: Arabic-First Experience

ID	Requirement
FR-101	The system shall render all UI strings, labels, error messages, and onboarding screens in Arabic when the user's device locale is set to Arabic or when the user manually selects Arabic from the language selector.
FR-102	The system shall include Arabic as a supported language in all Gemini API system prompts, ensuring recipe conversions (F3), snack suggestions (F4), and food recognition outputs (F2) return Arabic-language responses when the user's language preference is Arabic.
FR-103	The system shall persist the user's language preference in the Firestore user profile document, and apply it consistently across all sessions without requiring re-selection.
FR-104	The system shall automatically mirror the UI layout direction to Right-to-Left (RTL) when Arabic is the active language, ensuring proper alignment of text, icons, navigation components, and charts.

2.2 US-02: Photo-Based Meal Logging

ID	Requirement
FR-201	The system shall provide a camera capture interface that accepts a food photograph and transmits the image to the Gemini Vision API for nutritional analysis, returning an itemized calorie and macro estimate.
FR-202	The system shall display the AI-generated food recognition result (food name, estimated calories, protein, carbohydrates, fat) in an editable confirmation screen before persisting the entry to Firestore.
FR-203	The system shall complete the full photo-to-confirmation flow (image capture → API call → result display) within a maximum of 5 seconds under standard network conditions (4G/Wi-Fi), as defined by the project KPIs.

2.3 US-03: Accurate & Correctable Nutritional Data

ID	Requirement
FR-301	The system shall allow the user to edit any field (food name, calories, protein, carbohydrates, fat) of an AI-generated estimate on the confirmation screen before saving to Firestore.
FR-302	The system shall support full CRUD operations on all food log entries stored in Firestore, enabling the user to create, read, update, and delete any previously saved meal entry at any time.
FR-303	The system shall support full CRUD operations on user profile data (weight, height, activity level, dietary goals) stored in Firestore, ensuring no data point is permanently immutable after entry.

2.4 US-04: Local & Arab Cuisine Awareness

ID	Requirement
FR-401	The system shall include Arab cuisine context (common dishes, regional ingredients, local preparation methods) in the Gemini API system prompt for the Recipe Converter feature (F3), enabling it to generate healthy alternatives for traditional Arab meals.
FR-402	The system shall include Arab snack and street food awareness in the Gemini API system prompt for the Snack Discovery feature (F4), so that snack suggestions are culturally relevant to the user's locale.

2.5 US-05: Rapid, Error-Free Onboarding

ID	Requirement
FR-501	The system shall implement an onboarding flow that collects user profile data (age, gender, weight, height, activity level, dietary goals) and computes BMR and TDEE using the Mifflin-St Jeor equation, completing the full flow in no more than 5 screens.
FR-502	The system shall validate all numeric inputs during onboarding (weight, height, age) in real time, rejecting non-numeric characters, out-of-range values, and malformed decimal separators, and displaying inline error messages in the user's selected language.
FR-503	The system shall support both metric (kg/cm) and imperial (lb/ft-in) units during onboarding, with correct bidirectional conversion logic covered by unit tests.

2.6 US-06: Information-Dense Progress Dashboard

ID	Requirement
FR-601	The system shall render a dashboard screen using <code>fl_chart</code> , <code>GridView</code> , and <code>Stack</code> widgets that displays the user's daily calorie intake (ring/gauge chart), macronutrient breakdown, and weight trend (line chart) in a single scrollable view.
FR-602	The system shall fetch and display the dashboard data from Firestore in real time, reflecting any newly logged meal or profile update without requiring a manual refresh or app restart.

2.7 US-07: Healthy Alternative Discovery

ID	Requirement
FR-701	The system shall send the user's dietary profile (caloric target, restrictions, preferences) as context in the Gemini Text API prompt when generating recipe alternatives via the Recipe Converter feature (F3), ensuring suggestions are personalized.
FR-702	The system shall provide a Snack Discovery interface (F4) that sends a search/discovery prompt to the Gemini Text API, returning a list of healthy snack options tailored to the user's dietary restrictions and daily remaining caloric budget.

2.8 US-08: Water Intake Tracking

ID	Requirement
FR-801	The system shall provide a water logging interface with predefined volume increments of 250 ml and 500 ml, as well as a custom input option, and persist all entries to Firestore with timestamps.
FR-802	The system shall derive a personalized daily hydration goal from the user's weight and display progress toward that goal on the dashboard using a fill gauge or progress ring.
FR-803	The system shall support full CRUD operations on water intake entries stored in Firestore, enabling the user to view, edit, and delete previously logged hydration records.

3 Use Case Diagram

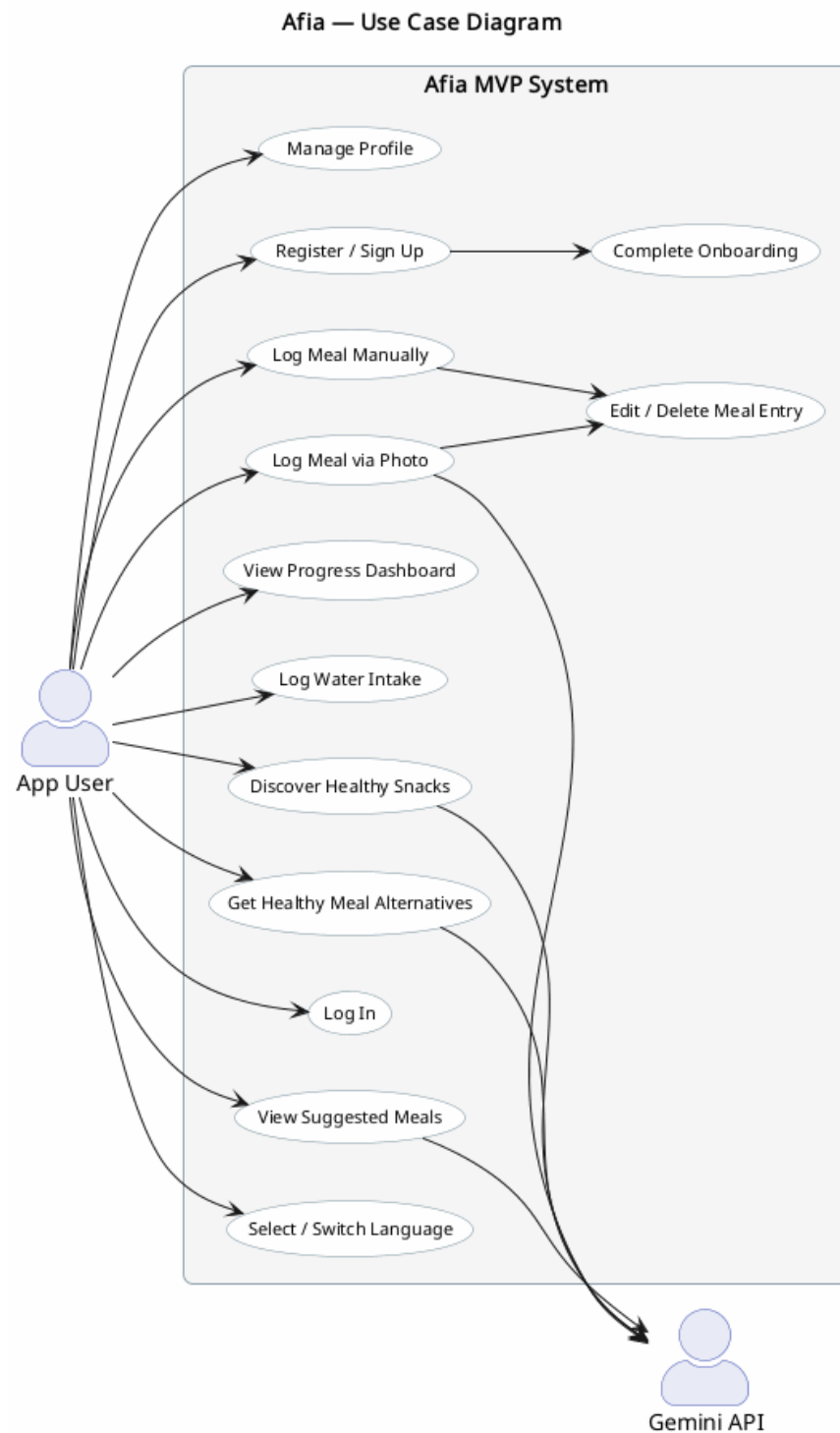
Figure 1 presents the system-level use case diagram for the Afia MVP. The **App User** is the sole primary (human) actor; the **Gemini API** appears as a secondary external actor that participates in AI-powered features. Firebase Authentication and Cloud Firestore are treated as internal infrastructure within the system boundary and are therefore omitted as actors.

4 Entity-Relationship Diagram

Figure 2 presents the ERD for the Afia MVP data model, including users, food items, meal and water logs, AI-generated suggestions, and saved favourites.

4.1 Actors

Actor	Type	Description
App User	Primary	The end user of the mobile application who logs meals, tracks water, views the dashboard, and manages their profile.
Gemini API	Secondary	Google’s generative-AI service used for photo-based food recognition (Vision) and text-based recipe/snack generation (Text).



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Figure 1: Afia — System Use Case Diagram

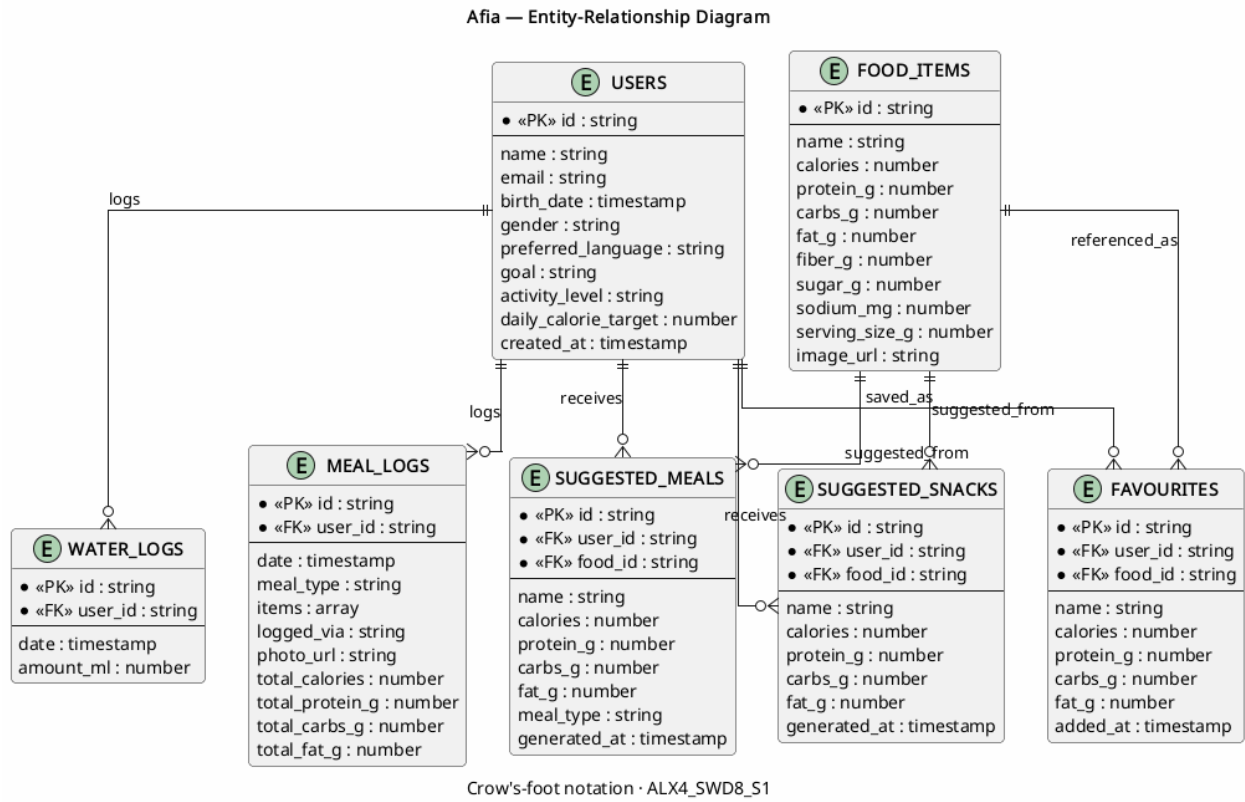


Figure 2: Afia — Entity-Relationship Diagram

4.2 Use Cases

#	Use Case	Description
1	Register / Sign Up	Create a new account via email and password through Firebase Authentication.
2	Complete Onboarding	Collect personal info, body metrics, activity level, and dietary goals across 5 screens; compute BMR/TDEE. <i>Included by</i> Register / Sign Up.
3	Log In	Authenticate with existing credentials via Firebase Authentication.
4	Log Meal via Photo	Capture a food photo and receive an AI-generated calorie/macro estimate from Gemini Vision.
5	Log Meal Manually	Enter meal details (food name, calories, macros) through a text-based form.
6	Edit / Delete Meal Entry	Modify or remove a previously logged meal entry. <i>Included by</i> Log Meal via Photo and Log Meal Manually.
7	Log Water Intake	Record water consumption using preset (250 ml / 500 ml) or custom volumes; view, edit, and delete entries.
8	View Progress Dashboard	Display calorie ring chart, macro breakdown, weight trend, and hydration progress in a single scrollable view.
9	Get Healthy Meal Alternatives	Submit a dish name and receive an AI-generated healthy recipe alternative via Gemini Text (Recipe Converter, F3).
10	Discover Healthy Snacks	Request personalized healthy snack suggestions via Gemini Text (Snack Discovery, F4).
11	Manage Profile	Update user profile data (weight, height, activity level, dietary goals) after onboarding.
12	Switch Language	Select the initial application language before authentication, or switch the language later from settings; preference persisted across sessions.

4.3 Relationships

- <<include>>: Register / Sign Up → Complete Onboarding
- <<include>>: Log Meal via Photo → Edit / Delete Meal Entry
- <<include>>: Log Meal Manually → Edit / Delete Meal Entry

4.4 Use Case to Functional Requirements Traceability

Use Case	Functional Requirements Covered
Register / Sign Up	NFR-S01
Complete Onboarding	FR-501, FR-502, FR-503
Log In	NFR-S01
Log Meal via Photo	FR-201, FR-202, FR-203
Log Meal Manually	FR-302 (create)
Edit / Delete Meal Entry	FR-301, FR-302 (update/delete)
Log Water Intake	FR-801, FR-802, FR-803
View Progress Dashboard	FR-601, FR-602, FR-802
Get Healthy Meal Alternatives	FR-401, FR-701
Discover Healthy Snacks	FR-402, FR-702
Manage Profile	FR-303
Switch Language	FR-101, FR-102, FR-103, FR-104

5 Non-Functional Requirements

5.1 Performance

ID	Requirement
NFR-P01	The system shall load the main dashboard screen (FR-601) and render all chart components within 2 seconds of navigation, measured from the point of screen transition to full widget render on a mid-range Android device (4 GB RAM, Snapdragon 600-series equivalent).
NFR-P02	The system shall gracefully handle Gemini API response times by displaying an active loading animation, and shall implement a strict 15-second timeout for all API requests to prevent the user interface from hanging indefinitely.
NFR-P03	The system shall implement local state-management caching (as defined in the risk mitigation plan) for frequent queries, preventing redundant API calls for identical prompts within the same session.
NFR-P04	The system shall gracefully handle failed API requests (e.g., timeouts or network errors) by displaying a clear, language-appropriate error message and providing a manual “Retry” action to the user, while maintaining a visible loading state during all active network operations.

5.2 Usability

ID	Requirement
NFR-U01	The onboarding flow (FR-501) shall be completable within 3 minutes by a first-time user, as validated during user testing sessions (Week 5, Milestone 5). This directly addresses the “First-Minute Failure” pain point.
NFR-U02	The photo-based meal logging flow (FR-201 → FR-202) shall require no more than 2 user interactions (capture + confirm) to complete a food log entry, reducing the competitor average of 5+ steps.
NFR-U03	All interactive elements shall meet a minimum touch target size of 48×48 dp, following Material Design accessibility guidelines.
NFR-U04	The system shall display inline, language-appropriate error messages for all form validation failures within 200 ms of user input.

5.3 Security & Data Integrity

ID	Requirement
NFR-S01	The system shall authenticate users via Firebase Authentication, supporting email/password sign-in, with all Firestore read/write operations protected by security rules scoped to the authenticated user’s UID.
NFR-S02	The system shall protect external API credentials by isolating the Gemini API key in a local environment configuration file (.env) excluded from version control, ensuring the key is not hardcoded in the repository source code while enforcing HTTPS for all network payloads.
NFR-S03	User profile and food log data stored in Cloud Firestore shall be accessible only to the owning user, enforced by Firestore security rules that deny all cross-user read/write access.

5.4 Reliability & Stability

ID	Requirement
NFR-R01	The system shall produce zero critical crashes during the MVP demo and testing phase, monitored via Firebase Crashlytics as specified in the project KPIs.
NFR-R02	Core dietary logic (BMR/TDEE calculators, unit conversions) shall maintain 100% unit-test coverage, with all core calculation functions individually tested, as specified in the project KPIs.

6 Traceability Matrix

User Story	Pain Point Solved	Functional Requirements	Non-Functional Requirements
US-01	Language Barrier	FR-101, FR-102, FR-103, FR-104	NFR-U04
US-02	Logging Friction	FR-201, FR-202, FR-203, FR-302, FR-701	NFR-P02, NFR-U02
US-03	Inaccurate Data & Correction Barrier	FR-301, FR-302, FR-303	NFR-S01, NFR-S03
US-04	Western-Centric Food Database	FR-401, FR-402	NFR-P02, NFR-P03
US-05	First-Minute Failure / Broken Onboarding	FR-501, FR-502, FR-503	NFR-U01, NFR-U03, NFR-U04, NFR-R02
US-06	UI Regression / Information Density Loss	FR-601, FR-602	NFR-P01
US-07	Lack of Actionable Guidance	FR-701, FR-702	NFR-P02, NFR-P03
US-08	Incomplete Health Tracking	FR-801, FR-802, FR-803	NFR-P01, NFR-S01, NFR-S03

6.1 Traceability to Proposal Features

Proposal Feature	Description	Related FRs
F1	Personalized caloric needs (BMR & TDEE)	FR-501, FR-503
F2	“Snap Your Plate” — AI Food Recognition	FR-201, FR-202, FR-203
F3	“Recipe Converter” — Healthy Alternatives	FR-401, FR-701
F4	Healthy Snack Discovery Prompts	FR-402, FR-702
F5	Interactive Progress Dashboard (f1_chart)	FR-601, FR-602
F6	Onboarding flow with diet & setup	FR-501, FR-502, FR-503
F7	Firebase Auth + Firestore Persistence	FR-103, FR-302, FR-303
F8	Manual override for AI calorie estimates	FR-301, FR-302
F9	(Future) RAG with verified food database	— (out of MVP scope)
F10	Water Intake Tracking	FR-801, FR-802, FR-803
F11	Bilingual Experience (Pre-auth selection & RTL)	FR-101, FR-102, FR-104

7 Future Scope Features

F9: RAG with Verified Food Database

Description: Building a Retrieval-Augmented Generation (RAG) backend connected to a vector database of verified, medically accurate Middle Eastern and Arab nutrition data to ensure AI estimates are mathematically consistent and highly accurate, eliminating AI hallucinations.

F12: Offline Mode & Data Synchronization

Description: Implementing local database caching (e.g., SQLite/Hive) and queuing mechanisms so users can view their dashboard, check past logs, and queue new meal entries without an internet connection. The system will automatically sync with Firestore once the network is restored.

F13: Clinical & Disease-Specific Dietary Management

Description: Expanding the user profile to include medical conditions (e.g., diabetes, hypertension, IBS) and severe allergies. The AI recommendation engine will cross-reference medical guidelines to

strictly filter out harmful ingredients and proactively suggest meals that safely support the user’s clinical profile.

F14: Automatic Retry with Exponential Backoff

Description: Implementing custom Dio interceptors to handle transient network failures autonomously. The system will use an exponential backoff algorithm (e.g., waiting 2s, 4s, 8s) to automatically retry failed Gemini API requests before surfacing an error to the user, ensuring a seamless experience on unstable mobile networks.

8 Appendix: Pain Point Evidence Summary

Pain Point	Competitor Signal Strength	Afia Resolution
No Arabic Support	MFP: 305, LoseIt: 57, Yazio: 36	Arabic-first UI, Arabic Gemini prompts, locale persistence (US-01)
Slow Logging	MFP: 347, LoseIt: 263, Yazio: 129	2-step photo logging via Gemini Vision (US-02)
Inaccurate Data	MFP: 347*, Yazio: 129*	AI estimates + manual override + full CRUD (US-03)
No Local Food	MFP: 305*, Yazio: 36*	Arab cuisine-aware prompt engineering (US-04)
Broken Onboarding	MFP: 53, LoseIt: 5, Yazio: 7	3-min KPI, unit-tested BMR/TDEE, input validation (US-05)
UI Density Loss	MFP: 105, LoseIt: 62, Yazio: 46	<code>fl_chart</code> + <code>GridView</code> , no decorative filler (US-06)
Passive Tracking Only	Qualitative (cross-app)	AI recipe + snack suggestions (US-07)
Incomplete Health Tracking	Qualitative (cross-app)	Water logging, hydration goals, and hydration CRUD (US-08)

Overlapping signal categories; value represents the parent theme signal.

This document is scoped exclusively to the 5-week MVP features (F1–F8, F10, F11) described in the project proposal. Features F9 (RAG with verified food database), F12 (offline mode & data synchronization), F13 (clinical & disease-specific dietary management), and F14 (automatic retry with exponential backoff) are acknowledged as future scope and excluded from functional requirements.